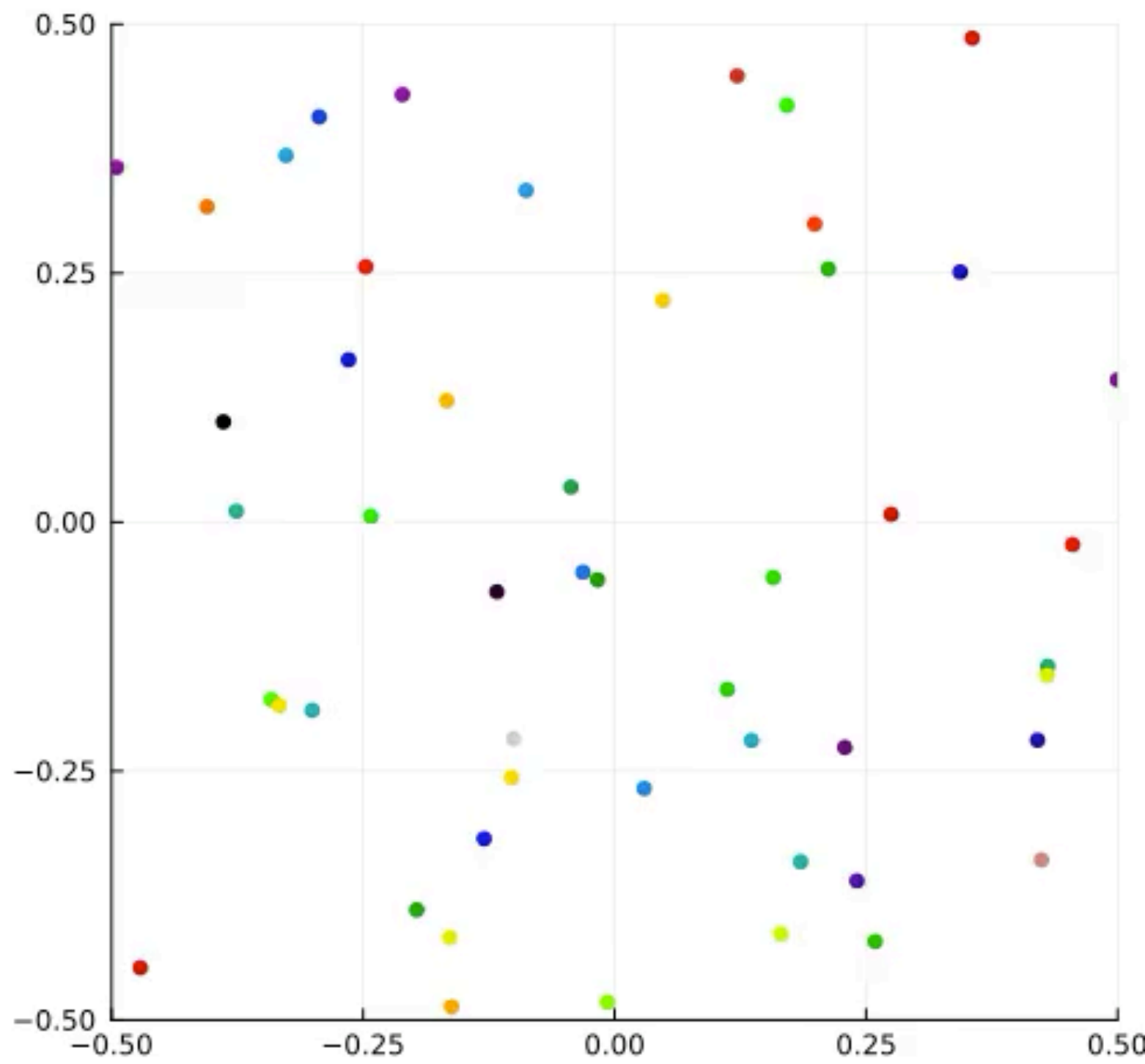


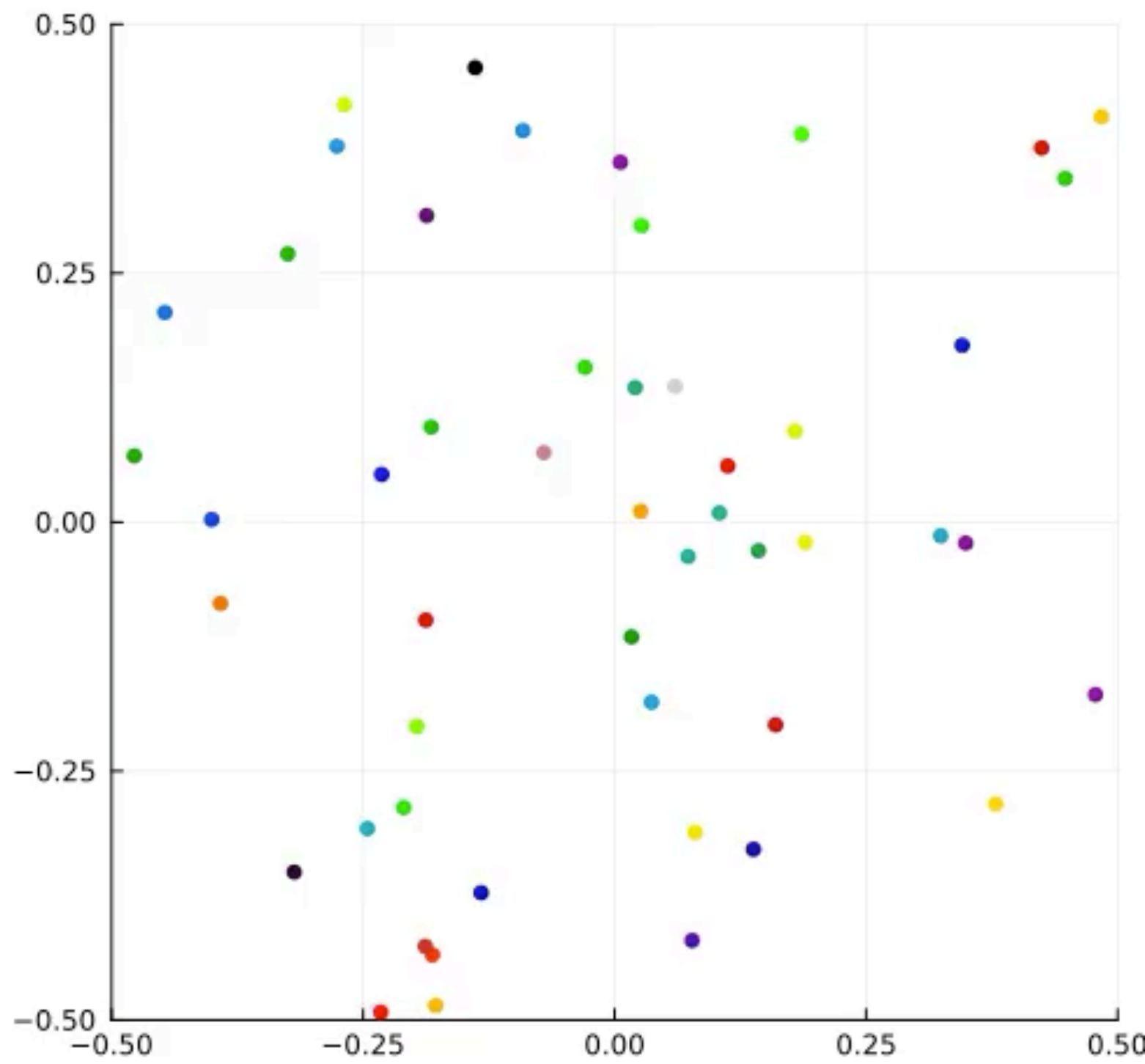
$$\bullet F = n(\theta_t^i) \cdot \frac{1}{N} \sum_{j \neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j) \text{ on } \mathbb{T}^2$$

$t=0.0$



$$\tau = 0$$

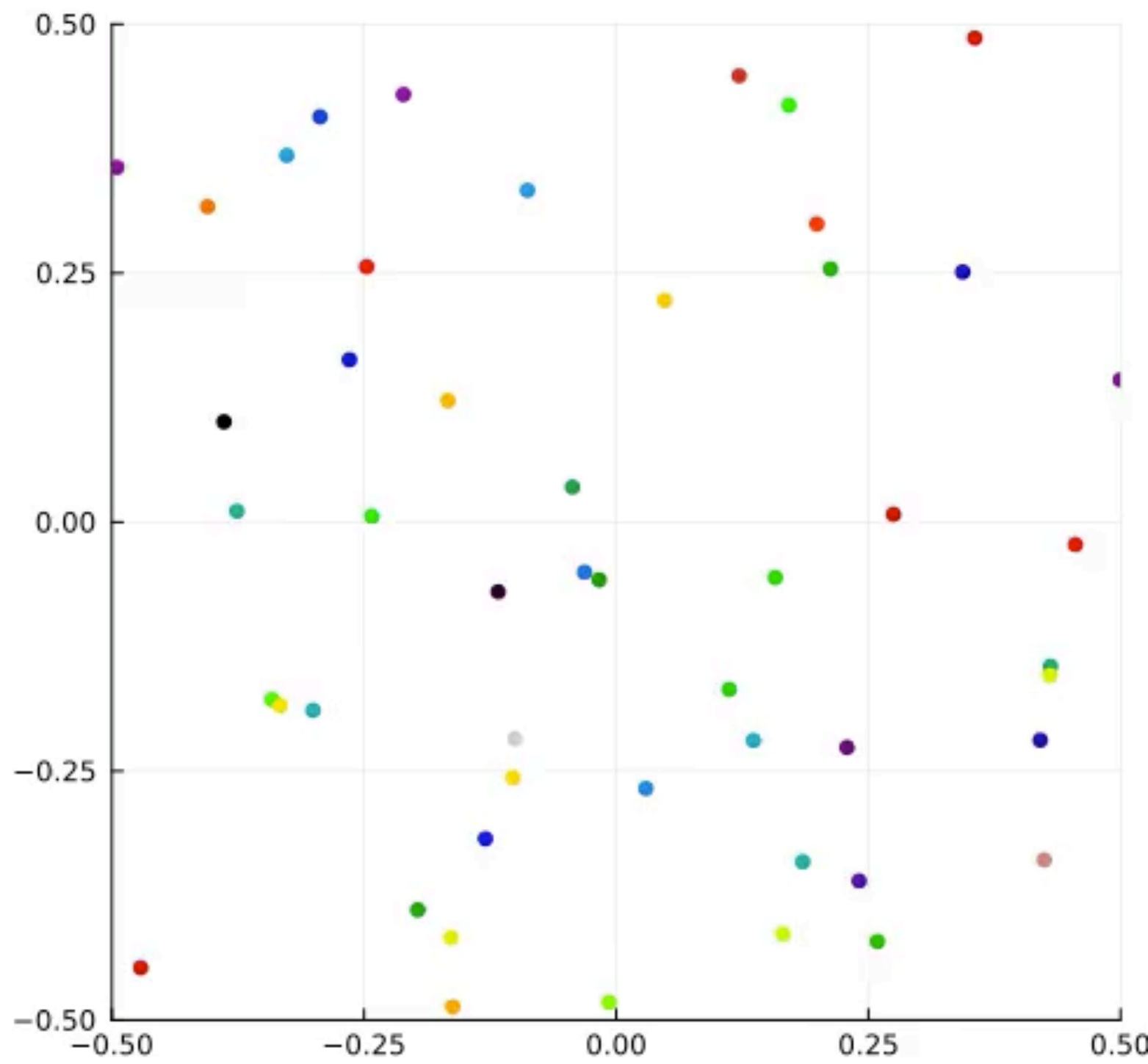
$t=0.0$



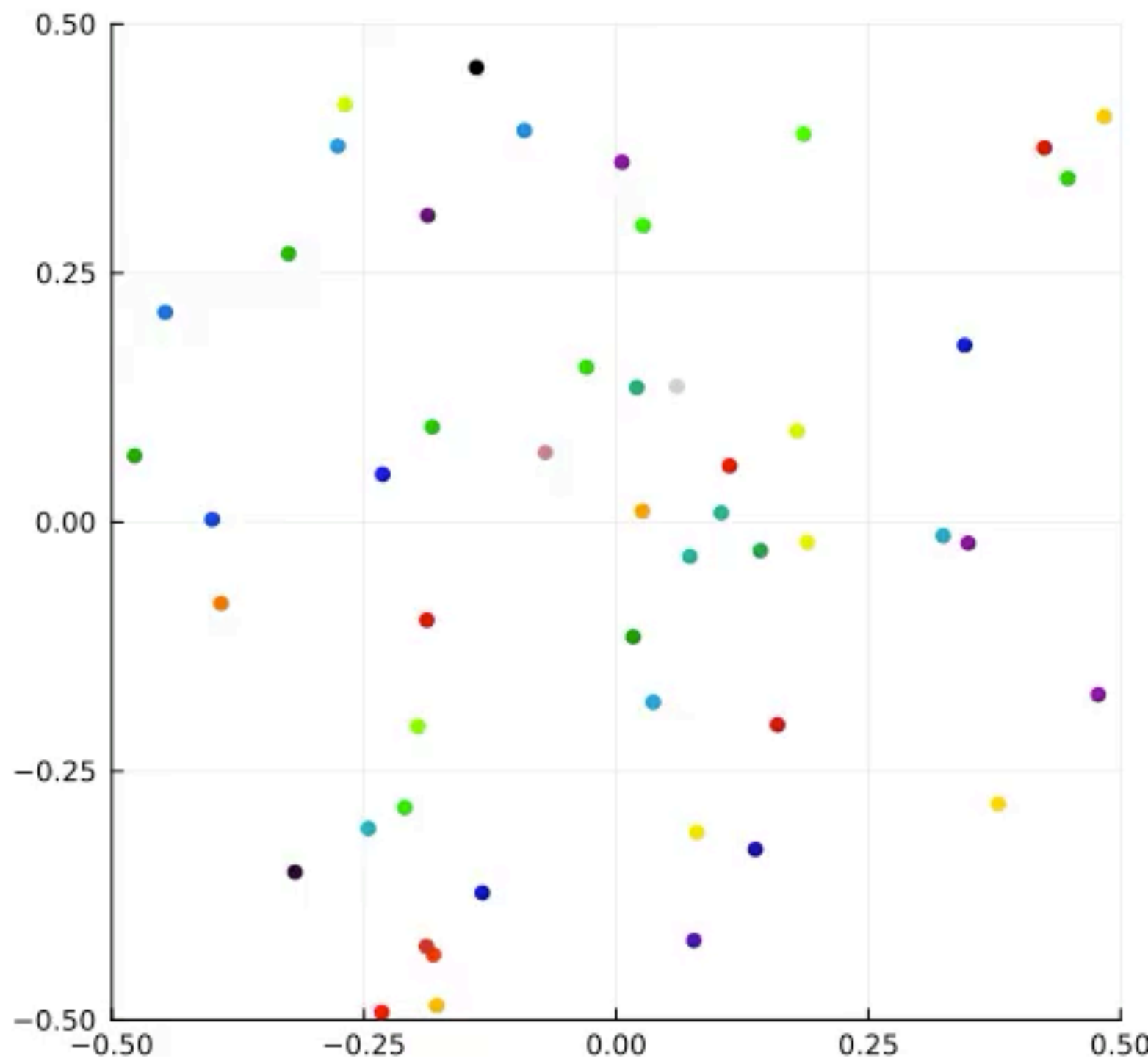
$$\tau = 0.15$$

Interacting ants

$t=0.0$

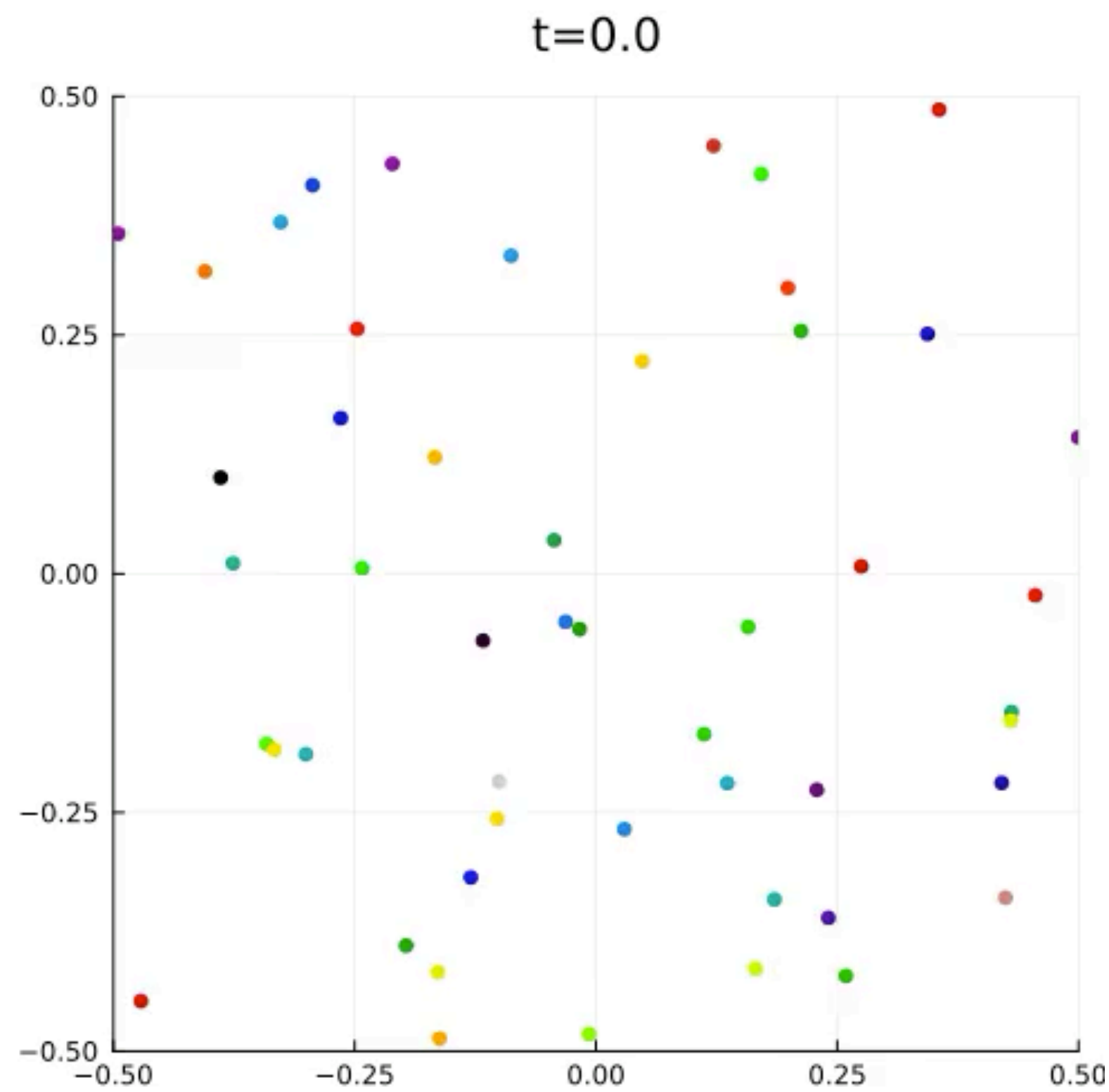


$t=0.0$

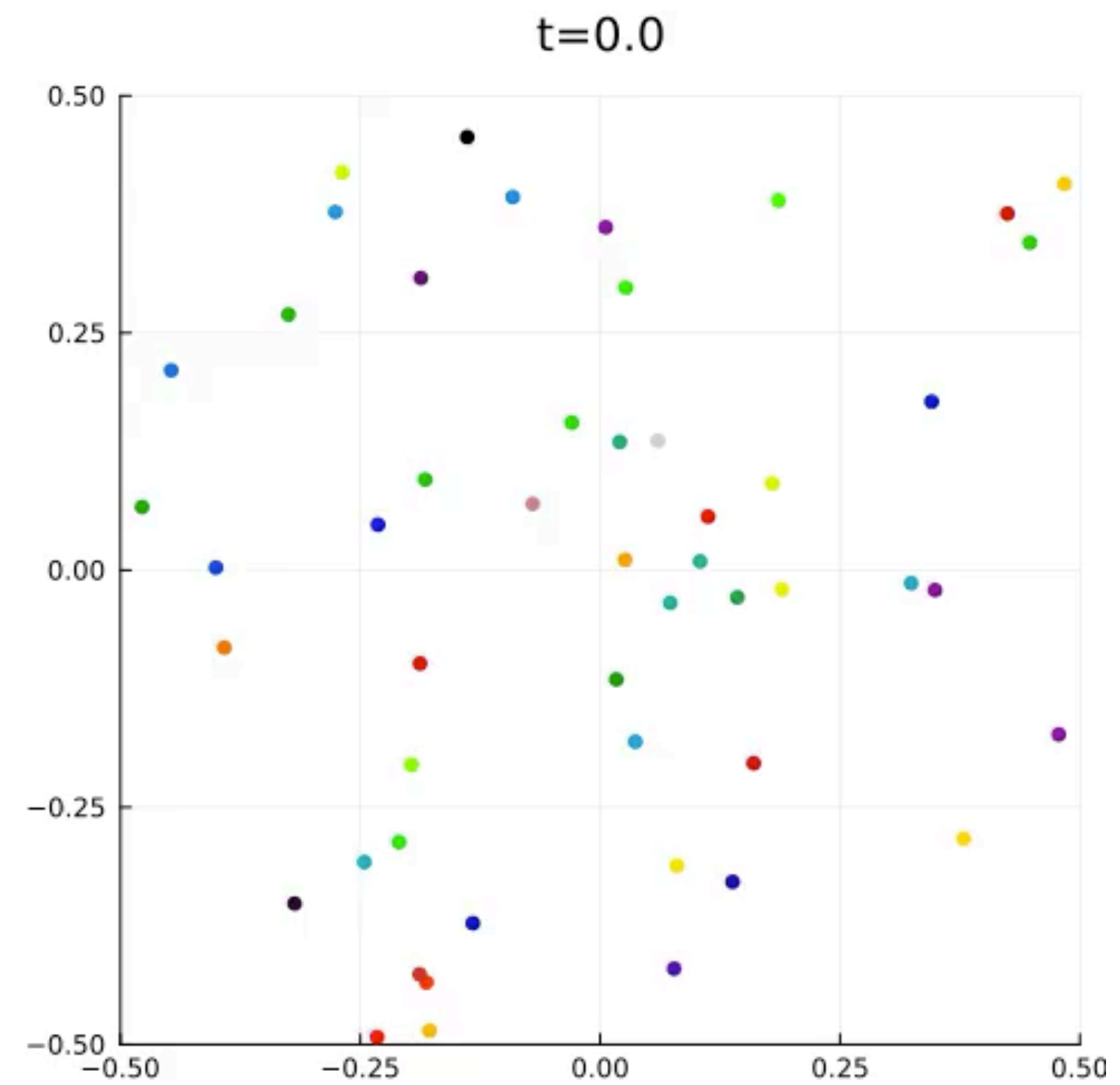


Interacting ants

- $F = n(\theta_t^i) \cdot \frac{1}{N} \sum_{j \neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j)$ on $\mathbb{T}^2$



$$\tau = 0$$



$$\tau = 0.15$$

Interacting ants

- $F = n(\theta_t^i) \cdot \frac{1}{N} \sum_{j \neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j)$ on $\mathbb{T}^2$
- Anticipation length τ as a phase transition/bifurcation parameter?